

João Rietra

Physician + AI Engineer · health AI, clinical validation and sensitive data

Recife, Brazil · +55 81 99243-3485 · CRM-PE 32980 (Brazilian medical licence)

joaorietra@gmail.com · jhlr.github.io · github.com/jhlr · linkedin.com/in/joaorietra

SUMMARY

AI Engineer who builds and validates AI models for healthcare. I measure calibration, out-of-distribution generalisation and explanation faithfulness as acceptance criteria before any model touches a clinical decision. I design decision-support rules from published consensus and test them the way the end user would use them. I handle sensitive data under Brazil's LGPD (GDPR-equivalent), with conditional access and an audit trail. I work with domain experts to turn clinical knowledge into a verifiable system.

KEYWORDS

Python · PyTorch · MLflow · Docker · AWS · Git · SQL · model calibration (ECE) · out-of-distribution evaluation (OOD) · LGPD and sensitive data · medical imaging (MedMNIST) · Grad-CAM and explainability · transfer learning · clinical validation · clinical decision support systems (CDSS) · clinical domain modelling · inferential statistics (CI, Wilcoxon) · dashboards (Streamlit) · fairness evaluation · domain-expert validation · technical mentoring and teaching · cross-functional teamwork

EDUCATION

- **Professional Master's in Software Engineering** — CESAR School, Recife · Oct 2025 – defence expected Nov 2026
 - **Doctor of Medicine (MD)** — FPS / IMIP, Recife · completed 2022 · CRM-PE 32980
 - **Science Without Borders exchange (Neuroscience)** — University of Alberta, Canada · 2015 – 2016
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EXPERIENCE

AI Engineer — LetzGrow (School Innovation, CESAR School) · 2026 – present

AI-assisted career platform with an AI mentor and a personalised development plan. Selected for 2026.2, as a two-person team.

- **Fairness evaluation** of the recommendation pipeline, reported in a versioned document.
- **Sensitive data:** conditional access and audit trail aligned with LGPD, on PostgreSQL 15.

ML Engineer — Veritas (School Innovation, CESAR School) · 2025 – 2026

Deepfake detection.

- **Explainability and generalisation:** Grad-CAM heatmap, false-positive analysis and out-of-distribution (OOD) evaluation, with contextual reading by a generative model.

Software Engineer (Clinical Domain + QA) — GasoGraph (Master's / CESAR) · 2025 – 2026

Clinical decision support system (CDSS) for arterial blood-gas interpretation in veterinary care (pH, PaCO₂, HCO₃⁻, BE, lactate), replacing printed tables and manual spreadsheets, with rules from published consensus.

- **Clinical decision graphs:** turned published consensus into flowcharts, from the measured value to the diagnostic hypothesis.
- **Rule engine:** implemented the rule tree from those flowcharts, from clinical design to code.
- **Clinically driven QA:** tested every version the way a physician would use it, checking that the decision reached the correct interpretation and that the tool met its clinical purpose.
- **Rule quality:** input validation (physiological ranges), handling of inconsistent values and test coverage on critical rules.

Teaching Assistant — CESAR School · 2025 – 2026

- Programming (C), Statistics (R), Calculus I, Competitive Programming and Data Analysis and Visualization.

Medical Data Analyst — J & L Hospital Consulting · 2023 – 2024

Analytics applied to hospital care and management, combining clinical and managerial data to support governance.

- **Clinical outcomes:** measured readmission, hospital-acquired infection and post-operative complications, pointing out failures and supporting protocol adjustments.
- **Quality and accreditation:** tracked the KPIs required by the ONA and JCI accreditation bodies.
- **Operations and costs:** analysed bed occupancy, operating room and inpatient costs, with Streamlit dashboards for hospital leadership.

Physician — Pediatric Emergency and Primary Care · 2022 – 2025

- Diagnostic decisions under high demand in a paediatric emergency department and longitudinal care in primary care.
- Practical grounding for clinical domain modelling, expert validation and handling sensitive data in AI solutions.

PROJECTS

Medical school thesis: diabetes in patients undergoing amputation

Retrospective observational study of 104 patients undergoing major lower-limb amputation (IMIP, 2011–2015): variable design, chart collection, statistical analysis (stepwise regression) and systematic comparison with national and international literature. Diabetes frequency of 80%, almost double that reported in comparable studies.

GlucoTrack · github.com/jhlr/GlucoTrack

Diabetes management app with blood-glucose and calorie logging.

ACADEMIC OUTPUT

1. *Training Reuse for Binary Triage in Multi-Domain Medical Imaging* (Master's dissertation, CESAR School, 2026): controlled comparison of 4 backbone origins across 5 domains (MedMNIST), evaluating AUC, calibration (ECE) and occlusion-based explanation faithfulness. Forensic intermediate fine-tuning won (AUC 0.842 vs. 0.796 for ImageNet) with far better calibration (ECE 0.045 vs. 0.219). MLflow and statistical aggregation (CI, Wilcoxon) over multiple seeds.
2. *Asymmetric generalisation in image-manipulation detectors: from the training domain to collapse on manual splicing* (SBTI 2026, submitted).
3. *Fine-tuning decision forests under drift: a Bayesian blend with a unified vocabulary* (SBTI 2026, submitted).
4. *EfficientNeXt: an autoregressive mini-LLM derived from the EfficientNet architecture with unified generation and embedding (GRIT) and decoding steering without an auxiliary model.*
5. *Automated inspection of electricity-meter reading photos: an OCR pipeline and characterisation of the coherence-validation problem* (SBTI 2026, submitted).

Papers written in Portuguese; original titles are in the Portuguese version of this CV.

SKILLS

Health AI: PyTorch · transfer learning and fine-tuning · medical imaging · Grad-CAM and explainability · calibration (ECE) · OOD evaluation · inferential statistics (CI, Wilcoxon) · clinical decision support systems (CDSS)

Engineering and data: Python · SQL · PostgreSQL · MLflow · Docker · AWS · Git · Streamlit · automated testing

Domain: clinical domain modelling · expert validation · LGPD and sensitive data · critical reading of literature and consensus

Languages: Portuguese (native) · English (native) · Spanish (basic) · French (basic)